# Translational Mechanical Systems

## (平动机械系统)

- Basic (Idealized) Modeling Elements (基本建模元件)
- Interconnection Relationships -Physical Laws (互联关系–物理规律)
- Derive Equation of Motion (EOM) – SDOF (推导运动方程–单自由度)
- Energy Transfer (能量转换)
- Series and Parallel Connections (串并联连接)
- Derive Equation of Motion (EOM) - MDOF (推导运动方程–多自由度)

# Key Concepts to Remember

- **Three Primary Elements of Interest (三个基本元件)**
  - Mass ( inertia ) $M$ 质量(惯量)
  - Stiffness ( spring ) $K$ 刚度(弹簧)
  - Dissipation ( damper ) $B$ 损耗(阻尼器)
    - *Distributed mass* (分布质量) → *lumped mass* (集中质量)
- **Lumped Parameters (集中参数)**
  - Mass maintains motion (质量维持运动) (Kinetic Energy (动能))
  - Stiffness restores motion (刚度储存运动) (Potential Energy (势能))
  - Damping eliminates motion (阻尼消除运动) (~~Eliminate Energy~~ (消除能量) ? )
    (Absorb Energy(吸收能量) )

# Variables

- $x$: **displacement** (位移) [m] ← **Reference Position (参考位置)& Assumed Positive Direction (假定正方向)**
- $v$: **velocity** (速度) [m/sec]
- $a$: **acceleration** (加速度) [m/sec$^2$]
- $f$: **force** [N] (力)
- $p$: **power** (功率) [Nm/sec]
- $w$: **work** (功) ( energy ) [Nm]

1 [Nm] = 1 [J] (Joule)

$$v = \dot{x} = \frac{d}{dt}x$$

$$a = \frac{dv}{dt} = \frac{d}{dt}\left(\frac{d}{dt}x\right) = \frac{d^2}{dt^2}x = \ddot{x}$$

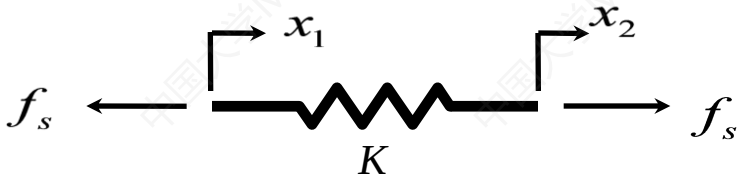
$$p = f \cdot v = f \cdot \dot{x} = \frac{d}{dt}w$$

$$w(t_1) = w(t_0) + \int_{t_0}^{t_1} p(t)\,dt$$

$$= w(t_0) + \int_{t_0}^{t_1} (f \cdot \dot{x})\,dt$$

# Basic (Idealized) Modeling Elements

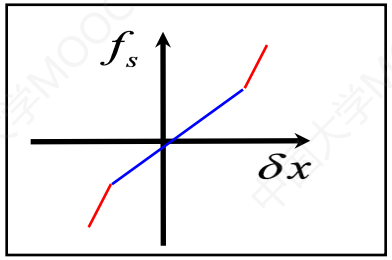
- **Spring (弹簧)**
  - Stiffness Element (刚度元件)



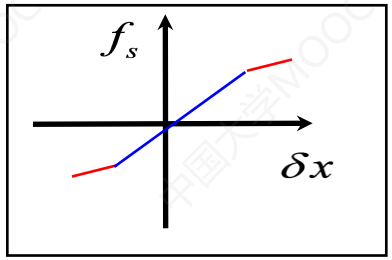

$$f_S = K\underbrace{(x_2 - x_1)}_{\delta x}$$

  - Idealization (理想)
    - Massless (无质量)
    - No Damping (无阻尼)
    - Linear (线性)
  - Stores Energy (储存能量)
    - Potential Energy (势能) $U = \frac{1}{2}k(\delta x)^2$

  - Reality (实际)
    - 1/3 of the spring mass may be considered into the lumped model (弹簧质量的1/3考虑到集中模型中).
    - In large displacement operation springs are *nonlinear* (大位移工作的弹簧是非线性的).




Hard Spring (硬弹簧)

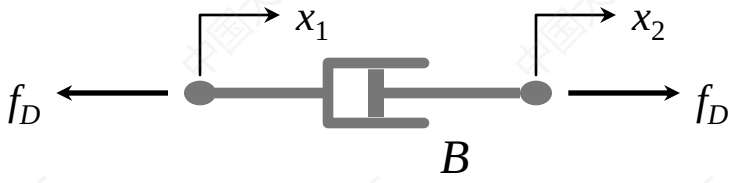



Soft Spring(软弹簧)

*Linear spring* (线性弹簧) ⇒
- *nonlinear spring* (非线性弹簧)
- *broken spring !!* (分段弹簧)

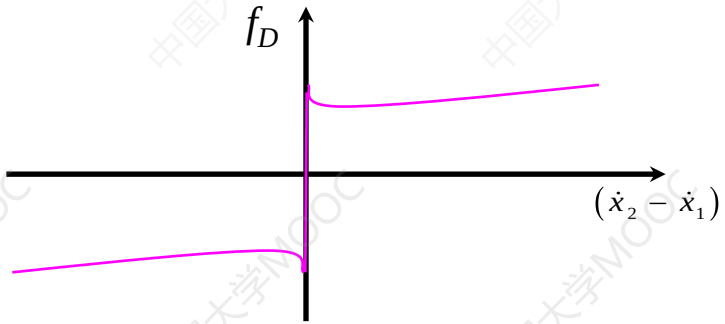
# Basic (Idealized) Modeling Elements

- **Damper** (阻尼器)
  - Friction Element (摩擦元件)

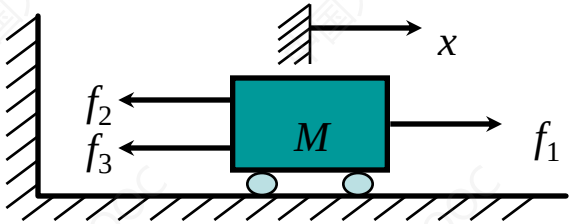



$$f_D = B(\dot{x}_2 - \dot{x}_1) = B(v_2 - v_1)$$

  - Dissipate Energy (耗散能量)




- **Mass** (质量)
  - Inertia Element (惯性元件)




$$M\ddot{x} = \sum_i f_i = f_1 - f_2 - f_3$$

  - Stores Kinetic Energy (存贮动能)

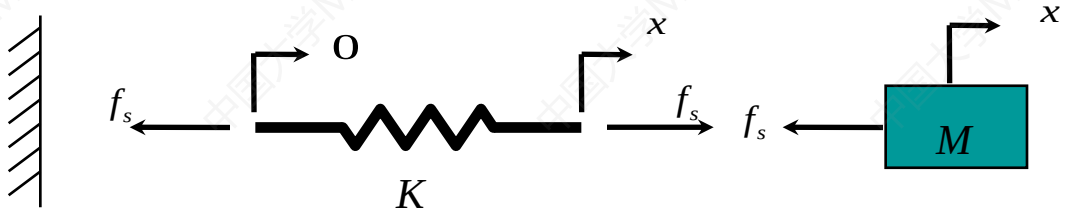
$$T = \frac{1}{2} M \dot{x}^2$$

# Interconnection Laws

- **Newton's Second Law (牛顿第二定律)**

$$\frac{d}{dt}(Mv) = M\ddot{x} = \sum_i f_{EXTi}$$

- **Newton's Third Law (牛顿第三定律)**
  - Action & Reaction Forces (作用力&反作用力)




$$f_s = K(x-0) = Kx$$

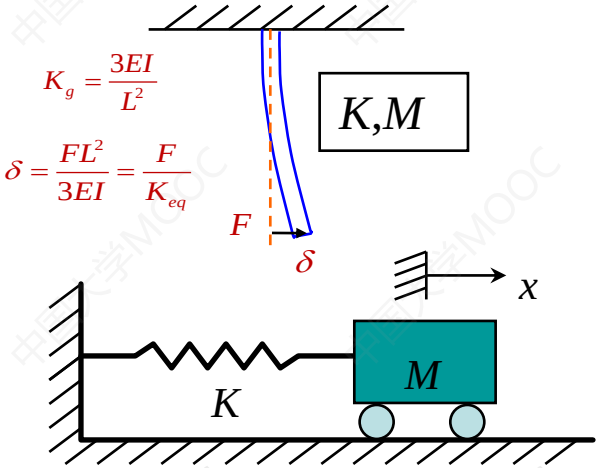
$$M\ddot{x} = -f_s$$

Massless spring (无质量弹簧)

$$M\ddot{x} + Kx = 0$$

E.O.M.

Ex: Lumped Model of a Flexible Beam (悬臂梁集中参数模型)

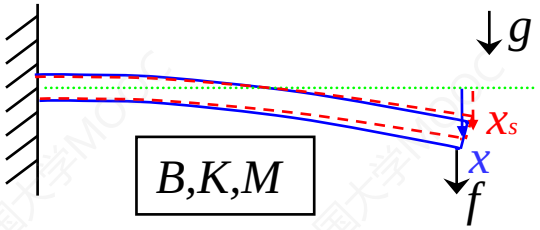


# Modeling Steps

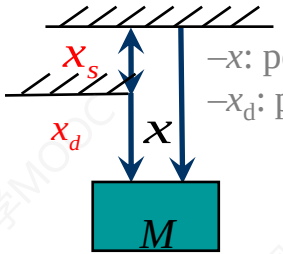
- **Understand System Function, Define Problem, and Identify Input/Output Variables** (理解系统功能、定义问题、确定输入输出量)
- **Draw Simplified Schematics Using Basic Elements** (画出由基本元件组成的系统简化原理图)
- **Develop Mathematical Model (Diff. Eq.)** (建立数学模型)
  - Identify reference point and positive direction (规定参考点和正方向)
  - Draw free-body-diagram (FBD) for each basic element (每个基本元件的受力分析)
  - Write elemental equations as well as Interconnecting Equations by applying physical laws. (*Check: # eq = # unk*) (运用物理规律写出元件方程和互联方程)
  - Combine equations by eliminating intermediate variables (消中间量合并方程)
- **Validate Model by Comparing Simulation Results with Physical Measurements** (通过仿真和实验结果的比较进行模型验证)

# Vertical Single Degree of Freedom (SDOF) System

- **Define Problem (定义问题):** **The motion of the object (物体运动)**
- **Input (输入量):** $f$, $Mg$
- **Output (输出量):** $x$
- **Develop Mathematical Model (Diff. Eq.) (列出数学模型)**




– Identify reference point and positive direction (定义参考点和正方向).

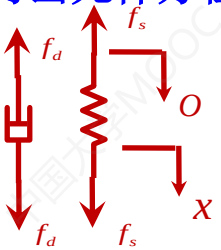



–$x$: position with no deflection of spring

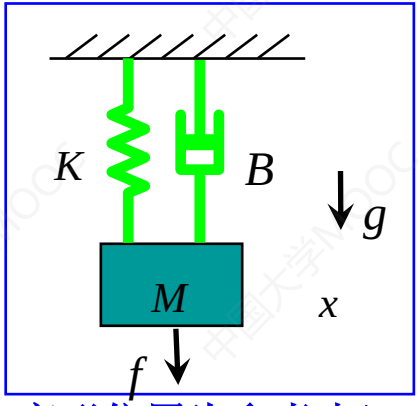
–$x_d$: position with a deflection of spring

$$x = x_s + x_d$$
$$K x_s = Mg$$

– Write Elemental Equations (写出元件方程)

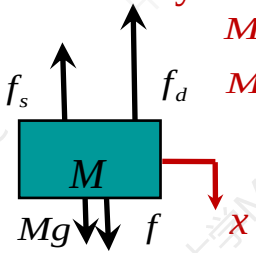



$$f_s = K(x - 0) = Kx$$
$$f_d = B\dot{x}$$




– Draw Free-Body-Diagram (FBD) (受力分析)

$$M\ddot{x} = Mg + f - f_s - f_d$$
$$Mg = Kx_s$$




- From the undeformed position (以未变形位置为参考点）

$$M\ddot{x} + B\dot{x} + Kx = Mg + f$$

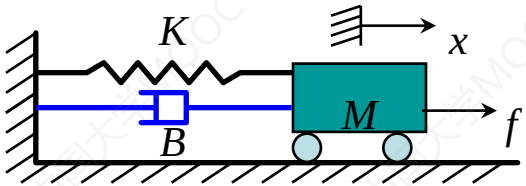
- From the deformed position (以变形(静平衡)位置为参考点)

$$M\ddot{x}_d + B\dot{x}_d + Kx_d = f$$

# Energy Distribution

- **EOM of a Simple Mass-Spring-Damper System (弹簧-质量-阻尼系统运动方程)**

$$\underbrace{M\ddot{x}}_{\substack{\text{Contribution}\\\text{of Inertia}}} + \underbrace{B\dot{x}}_{\substack{\text{Contribution}\\\text{of the Damper}}} + \underbrace{Kx}_{\substack{\text{Contribution}\\\text{of the Spring}}} = \underbrace{f(t)}_{\substack{\text{Total}\\\text{Applied Force}}}$$




*We want to look at the energy distribution of the system* (观察系统能量分配情况).

- **Multiply the Above Equation by the Velocity Term (方程两边乘以速度)**

$$M\ddot{x}\cdot\dot{x} + B\dot{x}\cdot\dot{x} + Kx\cdot\dot{x} = f(t)\cdot\dot{x}$$

- **Integrate the Second Equation w.r.t. Time (对时间积分)**

$$\underbrace{\int_{t_0}^{t_1} M\ddot{x}\cdot\dot{x}\,dt}_{\Delta KE} + \underbrace{\int_{t_0}^{t_1} B\dot{x}\cdot\dot{x}\,dt}_{\int_{t_0}^{t_1} B\dot{x}^2 dt \,\geq\, 0} + \underbrace{\int_{t_0}^{t_1} Kx\cdot\dot{x}\,dt}_{\Delta PE} = \underbrace{\int_{t_0}^{t_1} f(t)\cdot v\,dt}_{W}$$

$\Downarrow \frac{1}{2}M\dot{x}^2$ **Change of kinetic energy (动能变化)**

$\Downarrow$ **Energy dissipated by damper (通过阻尼能量耗散)**

$\Downarrow \frac{1}{2}Kx^2$ **Change of potential energy (势能变化)**

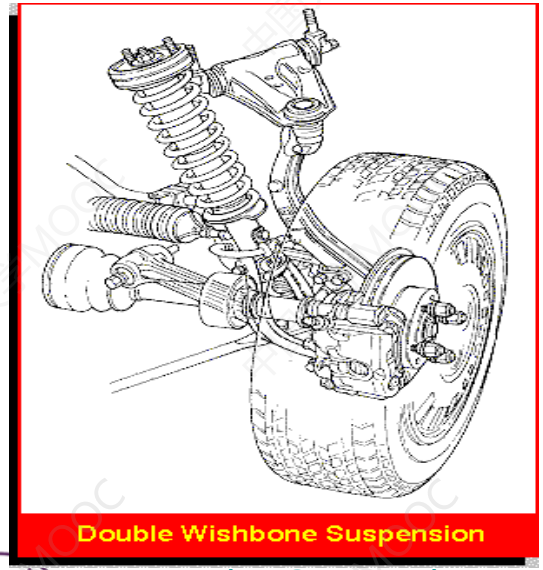
$\Downarrow$ **Total work done by applied force $f(t)$ from $t_0$ to $t_1$ (总做功量)**

# Example -- SDOF Suspension (Example)

- **Suspension System (悬挂系统)** – **Simplified Schematic (neglecting tire model) (简化的原理图(忽略轮胎模型))**

  Minimize the effect of the surface roughness of the road on the drivers comfort

  (最小化路面不平度对驾驶舒适性的影响)




Bose Active Suspension Control System Video

$$M\ddot{x} + B\left(\dot{x} - \dot{x}_p\right) + K\left(x - x_p\right) = -Mg$$

*From the "absolute zero"(绝对零位置)*

$$M\ddot{x}_{op} + B\dot{x}_{op} + Kx_{op} = -Mg - M\ddot{x}_p$$

*From the path (路径)*

$$x_{op} = x - x_p$$

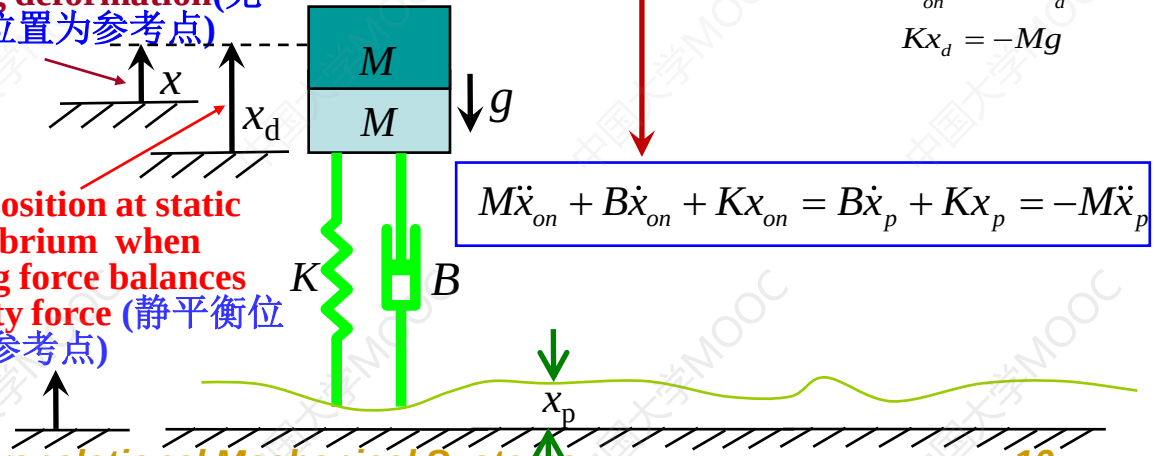
$$M\ddot{x}_{on} + B(\dot{x}_{on} - \dot{x}_p) + K(x_{on} - x_p) = 0$$

*From static position (静态位置)*

$$x_{on} = x - x_d$$

$$Kx_d = -Mg$$

**The position when no spring deformation(无变形位置为参考点)**

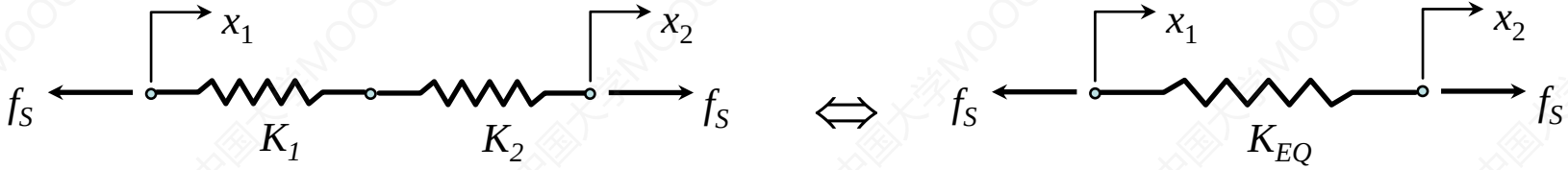


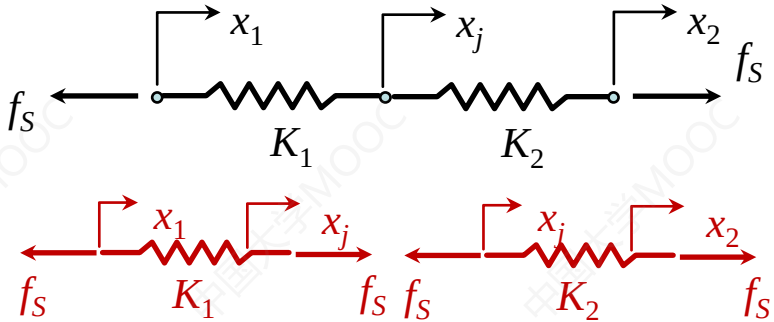

**The position at static equilibrium when spring force balances gravity force (静平衡位置为参考点)**

$$M\ddot{x}_{on} + B\dot{x}_{on} + Kx_{on} = B\dot{x}_p + Kx_p = -M\ddot{x}_p$$

# Series Connection

- **Springs in Series (串联弹簧)**




$$f_s = K_1(x_j - x_1)$$

$$f_s = K_2(x_2 - x_j)$$

$$K_1(x_j - x_1) = K_2(x_2 - x_j)$$

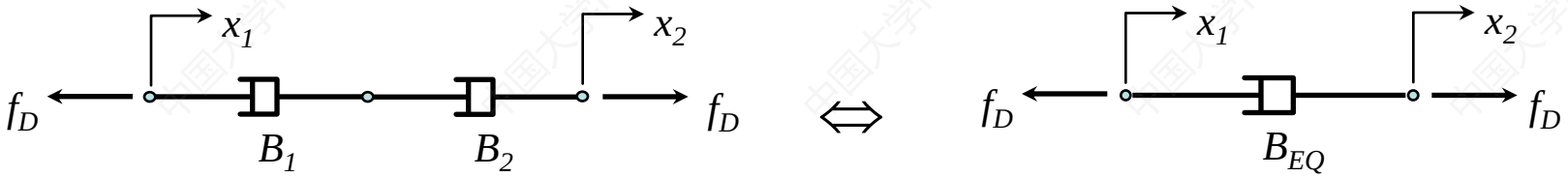
$$x_j = \frac{1}{K_1 + K_2}[K_2 x_2 + K_1 x_1]$$

$$f_s = K_1(x_j - x_1) = K_1\left\{\underbrace{\frac{1}{K_1 + K_2}[K_2 x_2 + K_1 x_1]}_{x_j} - x_1\right\}$$

$$\boxed{f_s = \underbrace{\frac{K_1 K_2}{K_1 + K_2}}_{K_{eq}}[x_2 - x_1]}$$

# Series Connection

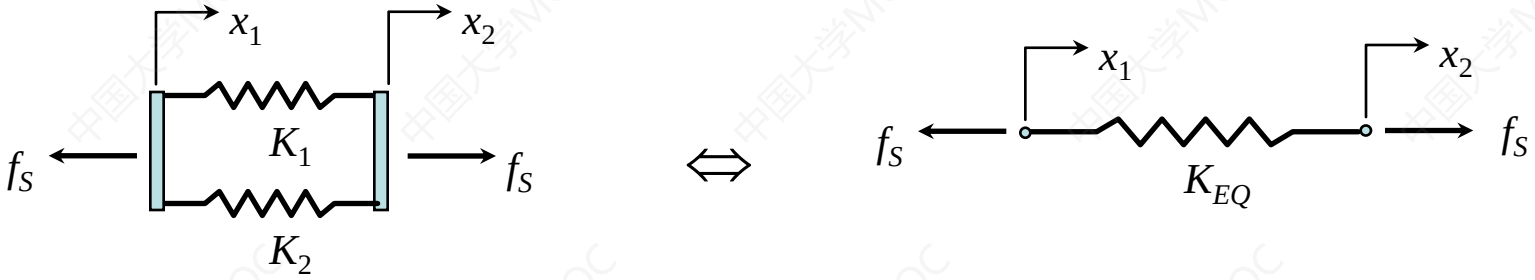
- **Dampers in Series (串联阻尼)**




$$f_d = \underbrace{\frac{B_1 B_2}{B_1 + B_2}}_{B_{eq}} [\dot{x}_2 - \dot{x}_1]$$

# Parallel Connection

- **Springs in Parallel** (并联弹簧)

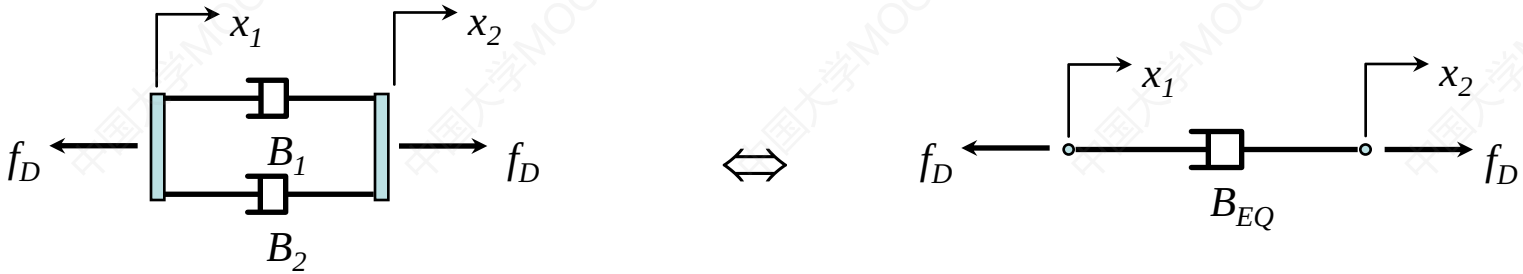



$$f_s = K_1(x_2 - x_1) + K_2(x_2 - x_1)$$

$$\boxed{f_s = \underbrace{(K_1 + K_2)}_{K_{eq}}(x_2 - x_1)}$$

# Parallel Connection

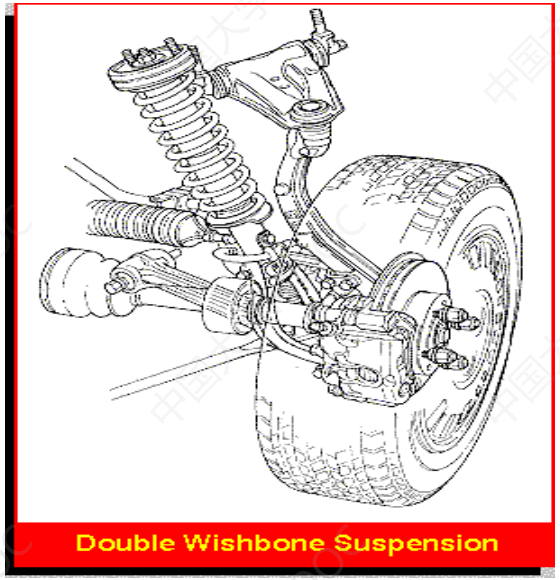
- **Dampers in Parallel** (并联阻尼)




$$f_D = \underbrace{(B_1 + B_2)}_{B_{EQ}}(\dot{x}_2 - \dot{x}_1)$$

# MDOF Suspension

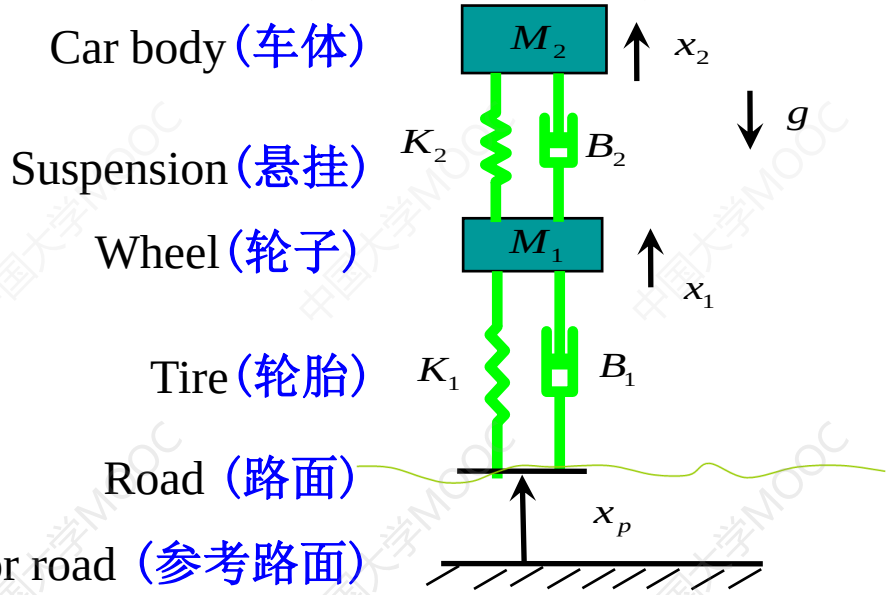
- **Suspension System**
  **(悬挂系统)**




– Simplified Schematic (with tire model)

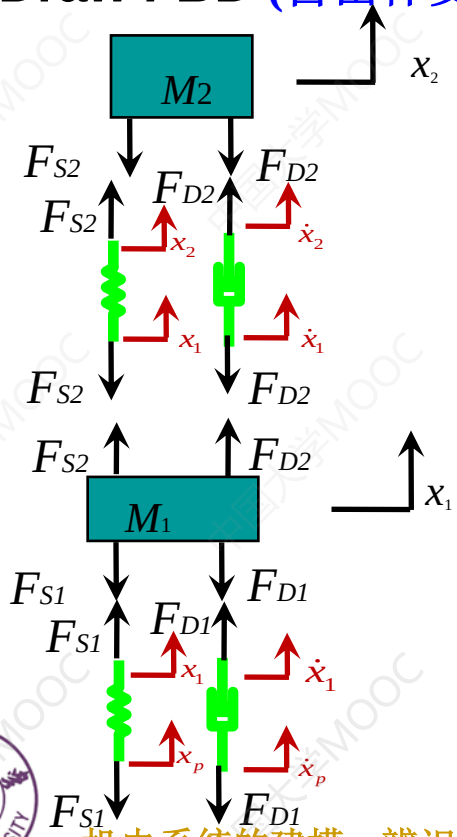
**(考虑轮胎模型的悬挂简化原理图)**

Assume refs. are defined to be static equilibrium positions under gravity (假定参考点为重力平衡下的静态平衡位置)



# Example — MDOF Suspension

– **Draw FBD** (自由体受力分析图)




– **Apply Interconnection Laws** (运用互联关系)

$$M_2\ddot{x}_2 = -F_{S2} - F_{D2}$$

$$F_{S2} = K_2\left(x_2 - x_1\right)$$

$$F_{D2} = B_2\left(\dot{x}_2 - \dot{x}_1\right)$$

$$M_1\ddot{x}_1 = F_{S2} + F_{D2} - F_{S1} - F_{D1}$$

$$F_{S1} = K_1\left(x_1 - x_p\right)$$

$$F_{D1} = B_1\left(\dot{x}_1 - \dot{x}_p\right)$$

$\dot{x}_p$ : is known from environment (来自环境影响)

UV: $x_2, F_{S2}, F_{D2}, x_1, F_{S1}, F_{D1}$

# Example — MDOF Suspension

– **Dynamics** (动力学方程)

$$M_2\ddot{x}_2 + B_2\dot{x}_2 - B_2\dot{x}_1 + K_2x_2 - K_2x_1 = 0$$

$$M_1\ddot{x}_1 - B_2\dot{x}_2 + \left(B_2 + B_1\right)\dot{x}_1 - K_2x_2 + \left(K_2 + K_1\right)x_1 = B_1\dot{x}_p + K_1x_p$$

– **Define Vector** (定义变量)

$$x = \begin{Bmatrix} x_2 \\ x_1 \end{Bmatrix}$$

– **Matrix Form** (矩阵形式的动力学方程)

$$\begin{bmatrix} M_2 & 0 \\ 0 & M_1 \end{bmatrix}\begin{Bmatrix} \ddot{x}_2 \\ \ddot{x}_1 \end{Bmatrix} + \begin{bmatrix} B_2 & -B_2 \\ -B_2 & B_2 + B_1 \end{bmatrix}\begin{Bmatrix} \dot{x}_2 \\ \dot{x}_1 \end{Bmatrix} + \begin{bmatrix} K_2 & -K_2 \\ -K_2 & K_2 + K_1 \end{bmatrix}\begin{Bmatrix} x_2 \\ x_1 \end{Bmatrix} = \begin{Bmatrix} 0 \\ B_1\dot{x}_p + K_1x_p \end{Bmatrix}$$

Mass matrix (质量矩阵)

Damping matrix (阻尼矩阵)

Stiffness matrix (刚度矩阵)

Input Vector (输入向量)